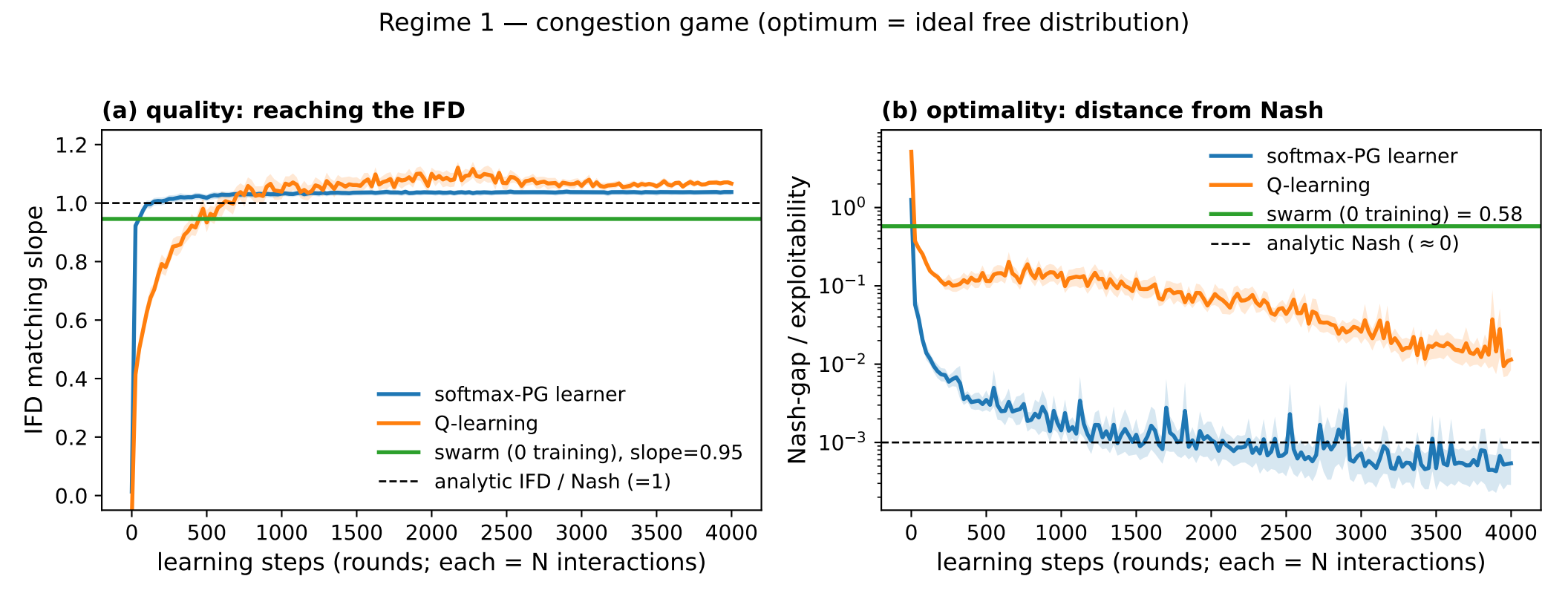
Regime 1 — congestion game (optimum = ideal free distribution)
(a) quality: reaching the IFD
IFD matching slope
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
2000
2500
3000
3500
4000
learning steps (rounds; each = N interactions)
softmax-PG learner
Q-learning
swarm (0 training), slope=0.95
analytic IFD / Nash (=1)
(b) optimality: distance from Nash
Nash-gap / exploitability
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
500
1000
1500
2000
2500
3000
3500
4000
learning steps (rounds; each = N interactions)
softmax-PG learner
Q-learning
swarm (0 training) = 0.58
analytic Nash (≈ 0)